

Analysis:

1) a) $q''_{g,abs} = 0.1067 \rightarrow 80 \text{ W/m}^2$
 $q''_{si,abs} = 0.856 \rightarrow 680 \text{ W/m}^2$

$$q''_{top} = (h + h_{rad})(T_{g,top} - T_{\infty})$$

$$R_a = \frac{0.0009 \text{ m}}{1 \frac{\text{W}}{\text{m}^2 \text{K}}} = 9 \times 10^{-4} \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R_{cond} = \frac{L_g}{k_g} + \frac{L_a}{k_a} + R_c = 0.0022 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R_{cond}' = \frac{1}{h} = 0.1 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R_{rad} = \frac{1}{h_{rad}} = 0.2 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R_c = 10^{-4} \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R_g = 0.0009 \text{ m} / 1 = 0.0009 \frac{\text{m}^2 \text{K}}{\text{W}}$$

b) $q''_{cond} = \frac{T_{si} - T_{g,top}}{R'_{cond}} = (1 - \eta)(0.856)$

$$0.1067 + q''_{cond} = (h + h_{rad})(T_{g,top} - T_{\infty})$$

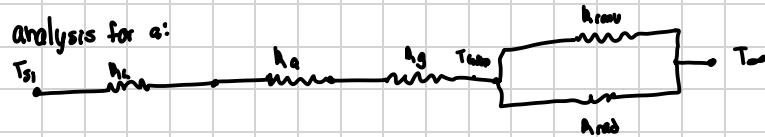
$$T_{si} = 342 \text{ K}$$

$$T_{g,top} = 341 \text{ K}$$

c) $\eta = a - b T_{si} = 0.21$

$$q''_{elec} = \eta (0.856) = 143 \text{ W/m}^2$$

analysis for a:



Given: $L_g = 3 \text{ mm}$

$$k_g = 1.5$$

$$L_a = 0.1 \text{ mm}$$

$$k_a = 1$$

$$T_{\infty} = T_{sur} = 300 \text{ K}$$

$$h = 10 \quad h_{rad} = 5 \text{ W/m}^2 \text{K}$$

$$\eta = a - b T_{si} \quad a = 0.353$$

$$b = 0.001$$

$$R_c = 10^{-4} \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$G = 800 \text{ W/m}^2$$

$$0.05 G \text{ reflect}$$

$$0.1067 \text{ absorbed top}$$

$$0.856 \text{ absorbed}$$

$$\text{silicon}$$

Find: a) resistance and network
 b) $T_{g,top}, T_{si}$
 c) η, q''_{elec}

Assumptions: steady state

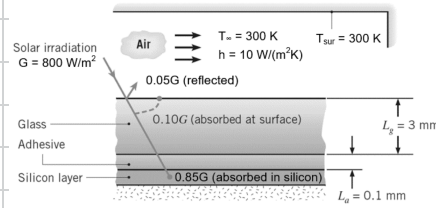
1D heat flow

constant k

$$q''_{back} = 0$$

uniform

Schematic:



2) (given: $D=16\text{mm}$ $k=80\text{ W/mK}$

$$T_w = 200^\circ\text{C}$$

$$L_{\text{ins}} = 260\text{mm}$$

$$T_\infty = 20^\circ\text{C} \quad h = 20\text{ W/m}^2\text{K}$$

$$T_{\text{max}} = 100^\circ\text{C} \text{ @ } T_0$$

Find:

a) T_0

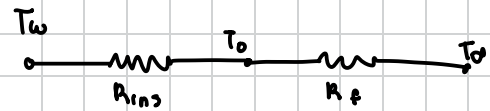
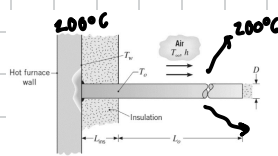
b) T_0 when $L_0 = 260\text{mm}$

c) L_0 so $T_0 \leq T_{\text{max}}$

Assumptions:

- steady state, constant air/material properties
- 1D conduction along rod
- neglect convection
- straight fin

Schematic:



Analysis:

$$a) \quad m = \sqrt{\frac{hP}{kAc}} \quad q_f = \sqrt{hPkAc} (T_0 - T_\infty) \tanh(mL_0)$$

$$q_{\text{cond}} = kAc \frac{T_w - T_0}{L_{\text{ins}}} = q_f$$

$$A = \frac{kAc}{L_{\text{ins}}} \quad B = \sqrt{hPkAc} \quad t = \tanh(mL_0)$$

$$T_0 = \frac{AT_w + (Bt)T_\infty}{A + Bt} = \frac{200 + 40 \tanh(10L_0)}{1 + 2 \tanh(10L_0)}$$

$$b) \quad T_0 = \frac{(6.03142)(200) + (0.0606)(20)}{0.03142 + 0.0606} = 81.5^\circ\text{C}$$

meets limits $T_0 < 100^\circ\text{C}$

$$c) \quad t = \frac{AL(T_w - T_{\text{max}})}{B(T_{\text{max}} - T_\infty)} = 0.625$$

$$mL_0 = \arctanh(t)$$

$$L_0 = \frac{\arctanh(0.625)}{10} = 73\text{mm}$$

3) Given:

$$k = 400 \text{ W/mK}$$

$$s = 10 \text{ mm} = 0.01 \text{ m}$$

$$h = 100 \text{ W/m}^2\text{K}$$

$$\eta_f = 0.6$$

convection tip

Find:

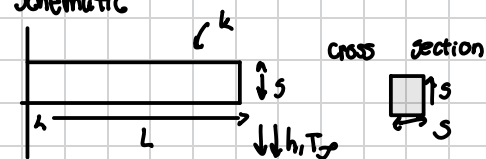
a) L

b) R_f, ϵ_f

Assumptions:

- steady state
- 1D fin conduction
- constant k, h
- convection
- uniform, no ΔE
- no radiation

schematic



a)

$$\Theta_b = T_b - T_\infty \quad M = \sqrt{hPkAc} \Theta_b$$

$$q_f = M \frac{\sinh(mL) + \beta \cosh(mL)}{\cosh(mL) + \beta \sinh(mL)}$$

$$\eta_f = \frac{q_f}{hA_f \Theta_b} = \frac{\sqrt{hPkAc}}{h(PL + A_t)} \frac{\sinh(mL) + \beta \cosh(mL)}{\cosh(mL) + \beta \sinh(mL)}$$

$$L = 146 \text{ mm}$$

$$Ac = 6^2 \quad P = 4s \quad A_t = Ac$$

$$A_f = PL + A_t$$

$$m = \sqrt{\frac{hP}{kAc}} = 10 \text{ m}^{-1} \quad \beta = \frac{h}{mk} = 0.025$$

$$Ac = 16^{-4} \quad P = 0.04$$

$$b) R_f = \frac{\Theta_b}{q_f} = \frac{1}{\eta_f h A_f} = 2.76 \text{ K/W}$$

$$C_f = \eta_f \frac{A_f}{Ac} = 35.9$$

$$L = 0.149 \text{ m}$$

$$A_f = PL + A_t = 0.00605 \text{ m}^2$$